Continuous Internal Evaluation – 1

1. Find all node running:

```
C:\hadoop-3.5.0\hadoop-3.5.0\sbin>jps
19408 DataNode
7424 ResourceManager
16900 NameNode
21572 Jps
21276 NodeManager
```

2. Store and Access the Dataset Using HDFS

Commands:

hdfs dfs -mkdir -p /bda/movie

```
C:\hadoop-3.5.0\hadoop-3.5.0\sbin>hdfs dfs -mkdir -p /bda/movie
```

hdfs dfs -put "C:\Users\satya\Downloads\movie_ratings_100.csv" /bda/movie/

hdfs dfs -ls /bda/movie

```
C:\hadoop-3.5.0\hadoop-3.5.0\sbin>hdfs dfs -ls /bda/movie
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::loadLibrary has been called by org.apache.hadoop.util.NativeCodeLoader
file:/C:/hadoop-3.5.0/hadoop-3.5.0/share/hadoop/common/hadoop-common-3.5.0.jar)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::arrayBaseOffset has been called by org.apache.hadoop.thirdparty.protobu
ssor (file:/C:/hadoop-3.5.0/hadoop-3.5.0/share/hadoop/common/lib/hadoop-shaded-protobuf_3_25-1.5.
WARNING: Please consider reporting this to the maintainers of class org.apache.hadoop.thirdparty.
oryAccessor
WARNING: sun.misc.Unsafe::arrayBaseOffset will be removed in a future release
Found 1 items
-rw-r--r--   1 satya supergroup       3967 2026-08-22 14:38 /bda/movie/movie_ratings_100.csv
```

3. Implement the Mapper:

Mapper.py

```
import sys
for line in sys.stdin:
    line = line.strip()
    # Skip header
    if line.startswith("user_id"):
        continue
    try:
        parts = line.split(",")
        user_id = parts[0]
```

```
        movie_id = parts[1]
        movie_title = parts[2]
        rating = float(parts[3])
        print(f"{movie_id}\t{movie_title}\t{rating}")
    except:
        continue
```

```
WARNING: sun.misc.Unsafe::arrayBaseOffset will be removed in a future release
101     The Last Horizon        10      40.00   4.00
102     Digital Dreams  10      44.00   4.40
103     Shadow Protocol 10      40.00   4.00
104     Beyond the Stars        10      41.00   4.10
105     Hidden Truth    10      39.00   3.90
106     Ocean of Time   10      45.00   4.50
107     Code Runner     10      36.00   3.60
108     The Final Quest 10      42.00   4.20
109     Parallel Worlds 10      44.00   4.40
110     Midnight Signal 10      38.00   3.80

C:\bda_movie>hdfs dfs -ls /bda/movie/output
```

4. Perform Distributed Aggregation

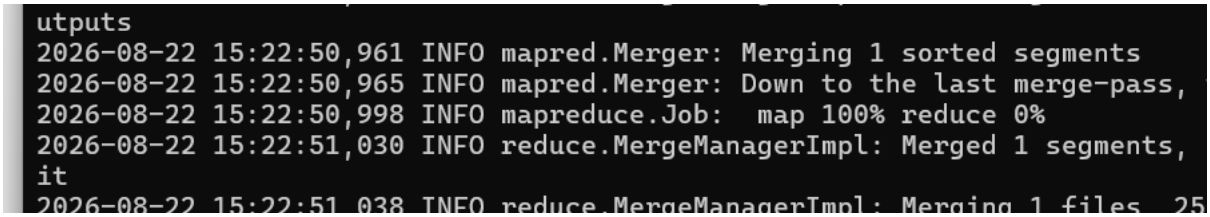
Reducer.py

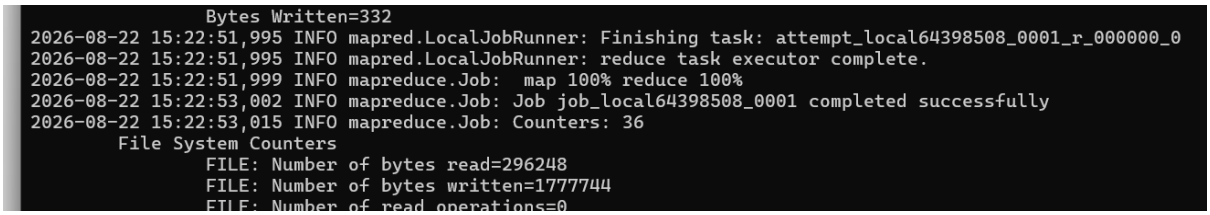
```
import sys
current_key = None
current_title = None
count = 0
total = 0.0
for line in sys.stdin:
    line = line.strip()
    if not line:
        continue
    parts = line.split("\t")
    if len(parts) != 2:
        continue
    key = parts[0]
    rating = float(parts[1])
    movie_id, movie_title = key.split("|", 1)
    if current_key != key:
        if current_key is not None:
            average = total / count
```

```
            # Task 4: Minimum 50 ratings
            if count >= 50:
                print(f"{current_key}\t{current_title}\t{count}\t{total:.2f}\t{average:.2f}")
        current_key = key
        current_title = movie_title
        count = 0
        total = 0.0
    count += 1
    total += rating
# Process last movie
if current_key is not None:
    average = total / count
    # Task 4: Minimum 50 ratings
    if count >= 50:
        print(f"{current_key}\t{current_title}\t{count}\t{total:.2f}\t{average:.2f}")
```

Hadoop MapReduce Execution:

hadoop jar C:\hadoop-3.5.0\hadoop-3.5.0\share\hadoop\tools\lib\hadoop-streaming-3.5.0.jar -files file:///C:/bda_movie/mapper.py -input /bda/movie/movie_ratings_100.csv -output /bda/movie/output -mapper "python mapper.py" -reducer "python reducer.py"





hdfs dfs -cat /bda/movie/output/part-00000

```
C:\hadoop-3.5.0\hadoop-3.5.0\sbin>hdfs dfs -cat /bda/movie/movie_ratings_100.csv
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::loadLibrary has been called by org.apache.hadoop.util.NativeCodeLoader in an unnamed module (
file:/C:/hadoop-3.5.0/hadoop-3.5.0/share/hadoop/common/hadoop-common-3.5.0.jar)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::arrayBaseOffset has been called by org.apache.hadoop.thirdparty.protobuf.UnsafeUtil$MemoryAcce
ssor (file:/C:/hadoop-3.5.0/hadoop-3.5.0/share/hadoop/common/lib/hadoop-shaded-protobuf_3_25-1.5.0.jar)
WARNING: Please consider reporting this to the maintainers of class org.apache.hadoop.thirdparty.protobuf.UnsafeUtil$Mem
oryAccessor
WARNING: sun.misc.Unsafe::arrayBaseOffset will be removed in a future release
user_id,movie_id,movie_title,rating,timestamp
U1007,101,The Last Horizon,5,2026-05-08
U1027,102,Digital Dreams,3,2026-02-19
U1008,103,Shadow Protocol,4,2026-02-07
U1007,104,Beyond the Stars,3,2026-04-23
U1140,105,Hidden Truth,5,2026-07-08
U1072,106,Ocean of Time,4,2026-01-25
U1109,107,Code Runner,5,2026-06-09
```

5. Apply a Minimum Rating Constraint

Modified reducer.py

Added

if count >= 50:

Output:

No records were produced because no movie has at least 50 ratings in given dataset.

6. Generate the Top 10 Movies

Top10.py

```
import sys
movies = []
for line in sys.stdin:
    line = line.strip()
    if not line:
        continue
    parts = line.split("\t")
    if len(parts) != 5:
        continue
    movie_id = parts[0]
    movie_title = parts[1]
    total_ratings = int(parts[2])
    average_rating = float(parts[4])
    movies.append((
        average_rating,
        total_ratings,
        movie_id,
        movie_title
```

```
    ))
movies.sort(key=lambda x: (-x[0], -x[1]))
for rank, movie in enumerate(movies[:10], 1):
    average_rating, total_ratings, movie_id, movie_title = movie
    print(
        f"{rank}\t{movie_id}\t{movie_title}\t"
        f"{average_rating:.2f}\t{total_ratings}"
    )
```

Execution Command:
python top10.py

output: No qualifying movies
Since Task 4 excluded all movies because each movie has fewer than 50 ratings, there are no qualifying movies available for Top-10 ranking.

7. Store and Display the Result

Store Output in HDFS:
hdfs dfs -mkdir -p /bda/movie/final_output
hdfs dfs -put final_result.txt /bda/movie/final_output/

Display: hdfs dfs -cat /bda/movie/final_output/final_result.txt

